Fundamentals of Computer Networks: LAN, WAN & PAN

Exploring Essential Local, Wide, and Personal Networks

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Contents

Networking Basics Overview 01

Comparing LAN, WAN & PAN 02

Wide Area Network (WAN) 03

Local Area Network (LAN) 04

Personal Area Network (PAN) 05

01

Networking Basics Overview

Introduction to Computer Networks

What is a Computer Network?

A computer network connects devices to share resources, exchange data, and enable communication, supporting tasks like file sharing, internet access, collaboration, and distributed applications.

Types, Scope, and Learning Path

Compare wired and wireless networks, introduce LAN, WAN, and PAN by coverage scope, and preview upcoming slides on definitions, features, pros and cons, use cases, and diagrams.

Network Classification by Scope

Local Area Network (LAN)

Covers a room, building, or campus. Usually owned by an organization; relies on Ethernet and Wi-Fi; moderate cost, configuration complexity, and internal management requirements.

Personal Area Network (PAN)

Very short range around a person, typically a few meters. Commonly owned by individuals, using Bluetooth or similar tech; low cost, simple setup, minimal management overhead.

Wide Area Network (WAN)

Spans cities, countries, or globally. Often owned or operated by telecom providers, using MPLS and carrier links; highest cost, complexity, and centralized management needs.

Core Networking Concepts for All Types

1 — Shared building blocks across all networks

All networks rely on common elements: end devices, interconnecting devices, transmission media, and protocols working together to enable reliable data communication across different environments.

2 — Devices, media, and protocols in practice

End devices connect via switches, routers, and access points over copper, fiber, or wireless links, using TCP/IP, Ethernet, Wi-Fi, and Bluetooth across LAN, WAN, and PAN scales.

Visual Map of LAN, WAN & PAN

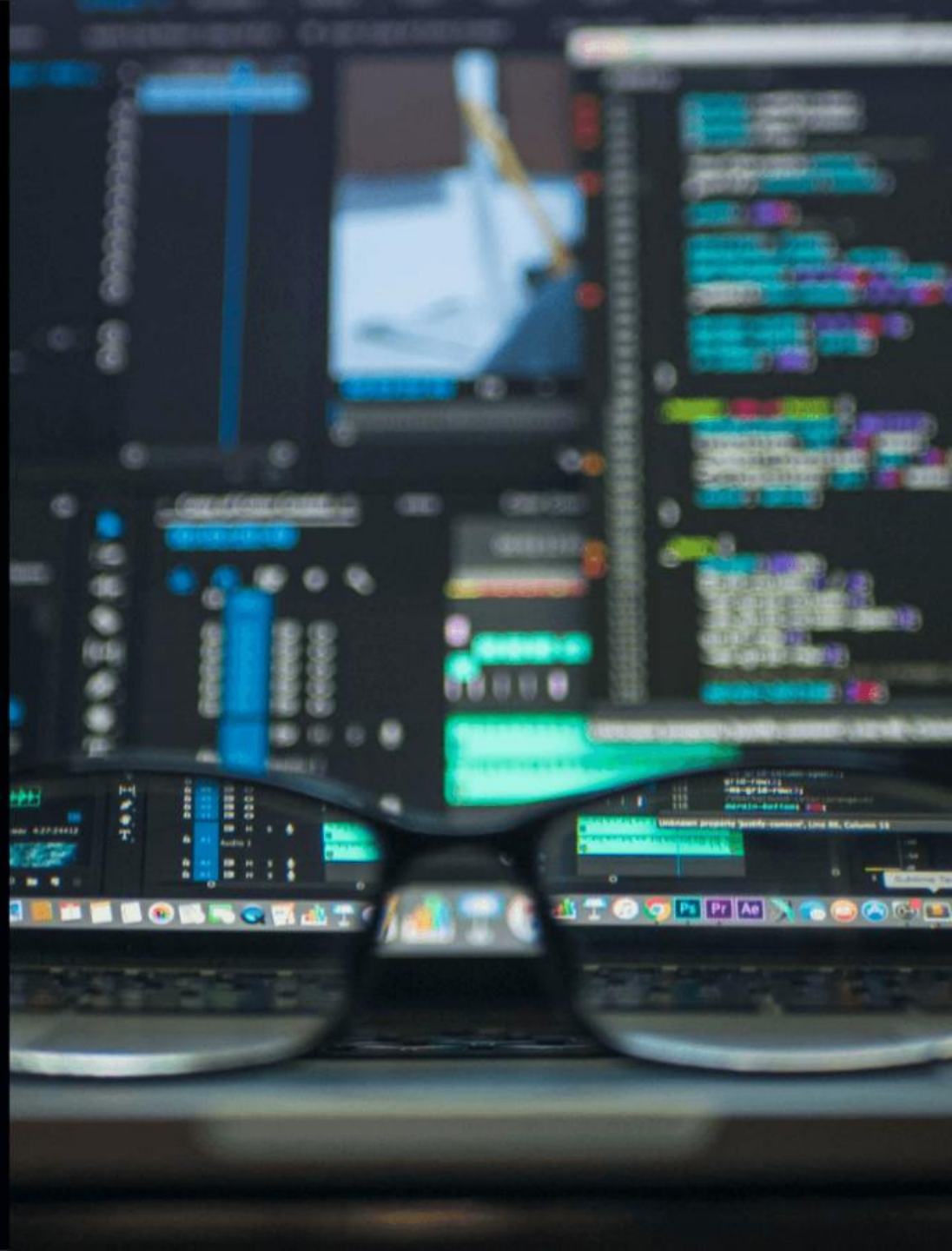
Layered Network Illustration

Central graphic shows a user surrounded by overlapping PAN, LAN, WAN circles, visually mapping how personal, local, and wide networks relate and build on each other.

PAN, LAN, and WAN Flow

Icons show PAN devices connecting to a LAN router, then multiple LANs linking through a WAN cloud, with arrows and captions explaining each step of data movement.

Customer Satisfaction	Recommendation rate
8.1	70%
8.2	71%
8.6	76%
7.9	69%
8.0	70%



02

Personal Area Network (PAN)

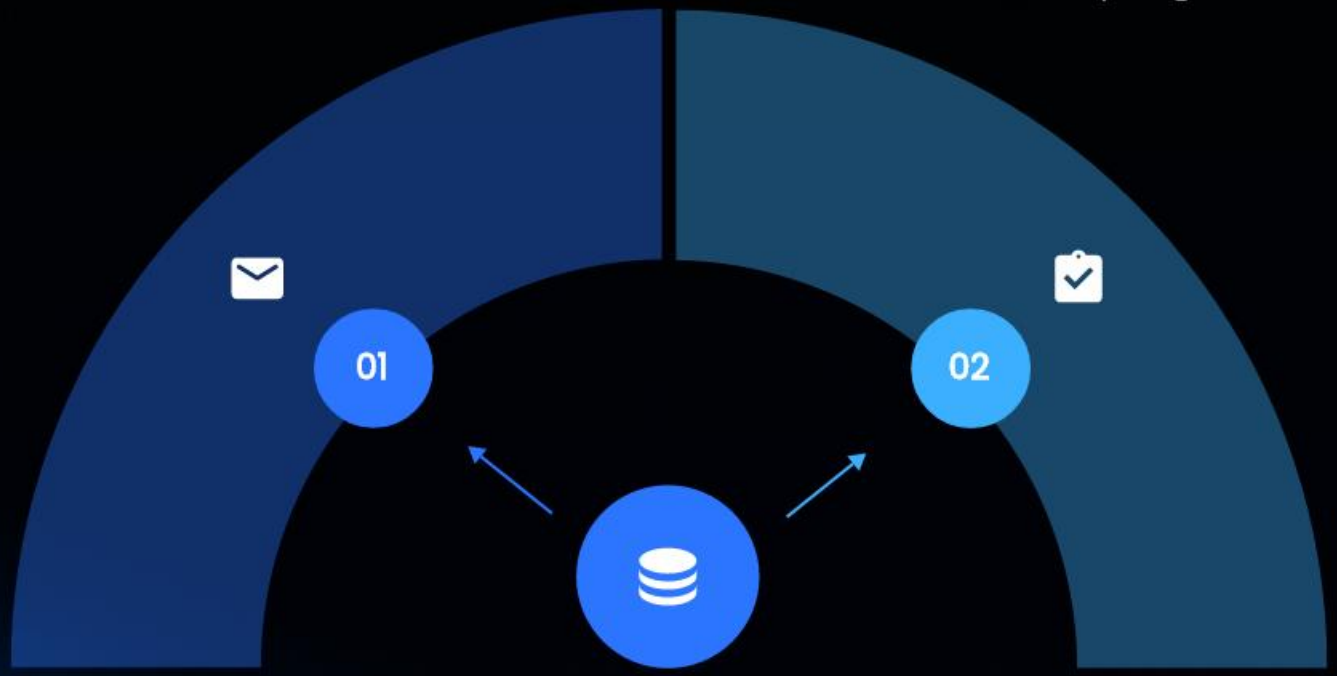
PAN Definition & Key Features

What is a Personal Area Network (PAN)?

A PAN is a very small network around one user or device, typically covering about 10 meters, enabling room, body, or workspace-level connectivity between personal digital devices.

Technologies, Devices, and Characteristics

PANs commonly use Bluetooth, IR, NFC, Zigbee, or USB tethering to link smartphones, wearables, earphones, and peripherals, emphasizing low power, short range, and simple ad-hoc or star topologies.



PAN Advantages & Disadvantages

Key Advantages of PAN

PANs offer convenient wire-free peripherals, simple setup, low cost, and low power usage, making them ideal for mobile users and wearable devices in everyday personal environments.

Key Disadvantages of PAN

PANs suffer from very limited range, low bandwidth unsuitable for heavy data, and potential interference and security risks, especially when multiple devices connect without secure pairing.

PAN Diagram & Typical Uses

Visual layout: PAN device ecosystem

Central smartphone connects via Bluetooth to smartwatch, earbuds, laptop, and car infotainment. Use icons, Bluetooth symbol, and short-range signal waves to emphasize limited physical coverage.

Typical PAN usage scenarios

Highlight personal audio with wireless headphones, fitness tracking via wearables, phone–laptop file sharing or tethering, and nearby smart home control devices operated through remotes or wearables.

03

Local Area Network (LAN)

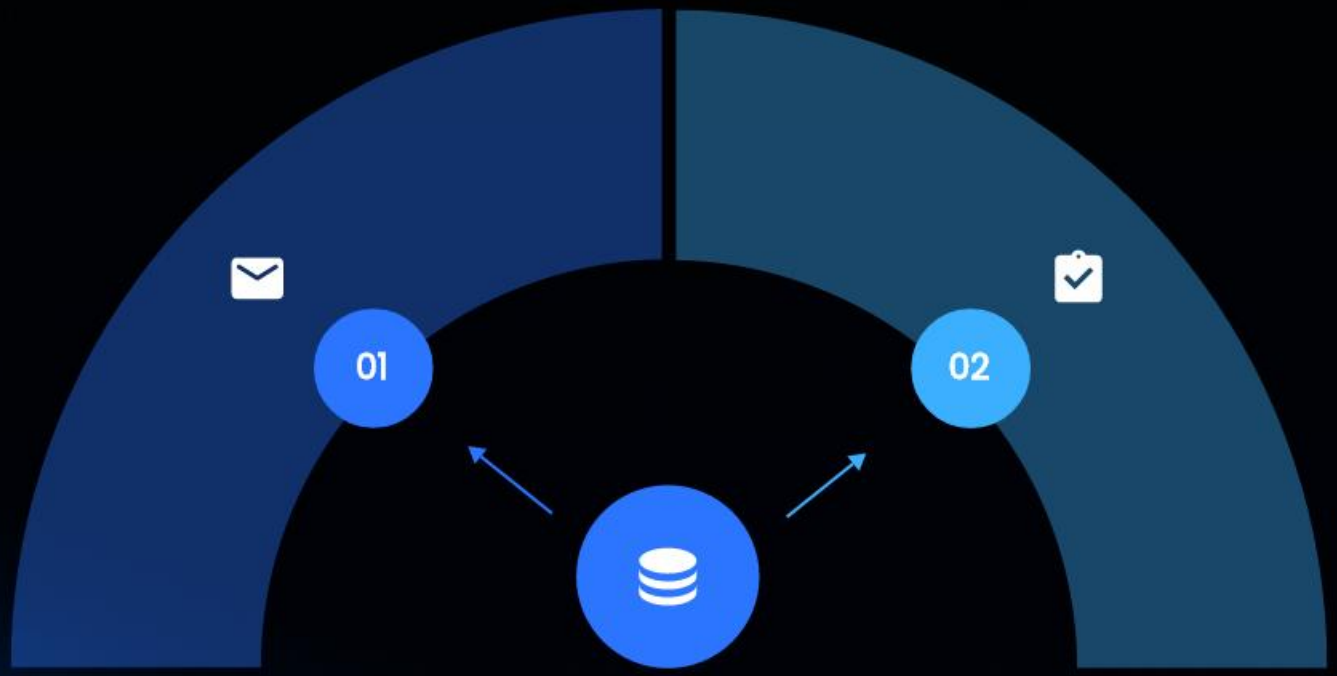
LAN Definition & Core Features

What Is a Local Area Network (LAN)?

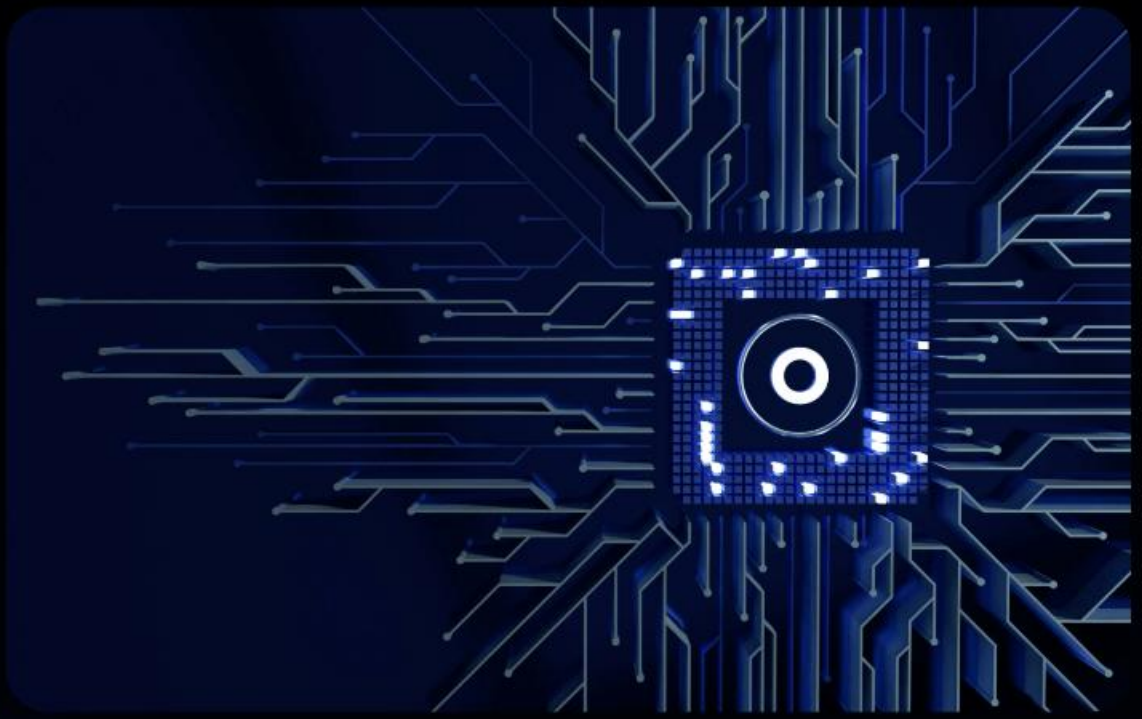
A LAN is a privately owned network covering a small geographic area, like a room, building, or campus, commonly used in homes, offices, and schools.

Key LAN Technologies and Characteristics

LANs use Ethernet, Wi-Fi, and sometimes fiber to connect routers, switches, access points, servers, PCs, and printers, providing high speed, low latency, and reliable communication.



LAN Topologies & Diagram



Common LAN Logical Topologies

Compare star, extended star, bus, and ring topologies. Highlight star as dominant today, bus as mostly historical, and ring as rare but still relevant in some specialized environments.

Star Layout & Example LAN Diagrams

Show central switch or router connecting PCs, laptops, printer, and server via Ethernet. Include Wi-Fi access point, home LAN diagram, and icons: switch, Wi-Fi router, Ethernet cable.

LAN Advantages & Disadvantages

Comparison Item A

Key Advantages of LAN

LANs offer high-speed data transfer for file sharing, streaming, and backups. They enable efficient sharing of printers, storage, and internet with centralized management and better security control.

VS

Comparison Item B

Key Disadvantages of LAN

LANs are geographically limited and incur setup and maintenance costs. Larger LANs demand configuration expertise and face internal security risks if access controls are weak.

LAN Applications & Use Case Overview

Home & Office LANs

Home LAN links personal devices for internet, media, and printer sharing. Office LAN connects employee PCs, file and database servers, IP phones, enabling collaboration, VoIP, and secure internal data access.

Campus, Business & Industrial LANs

Campus LAN supports labs, libraries, e-learning, campus Wi-Fi. Small business LAN connects POS, cameras, backups. Industrial LAN links machines, sensors, controllers, often integrated into wider WANs.

04

Wide Area Network (WAN)

WAN Definition & Characteristics

A Wide Area Network spans large geographic regions, interconnecting multiple LANs and PANs across cities, countries, or continents using routers and long-distance communication links.

WANs often combine telecom carrier infrastructure and enterprise networks, using leased lines, MPLS, VPNs, fiber, satellite, or 4G/5G, with wide coverage, varied bandwidth, higher latency, and complex management.

WAN Topology & Diagram

Typical WAN Traffic Paths Explained

Labeled paths show traffic flows: branch office to headquarters, headquarters to data center, and onward to cloud services, highlighting multi-hop routing across diverse network segments. Diagram emphasizes logical topology over physical distance, helping students understand how heterogeneous access technologies still form a unified wide area network infrastructure.

Visual Overview of WAN Connections

World map background with LAN sites—offices, campuses, data centers—interconnected through a central cloud-shaped WAN, illustrating global reach and multi-site enterprise networking. Routers at each local site connect into the WAN cloud, representing ISP or carrier networks; icons depict fiber, microwave, satellite, and undersea cable links across regions.

WAN Advantages & Disadvantages

Advantages of Wide Area Networks

WANs enable global connectivity, centralize data and services, and support remote work and business continuity across regions, improving collaboration, scalability, and access to shared organizational resources.

WANs allow organizations to interconnect branch offices, data centers, and cloud services, providing unified communication, centralized management, and consistent user experience across geographically dispersed locations.

Vs

Disadvantages and Challenges of WANs

WANs involve higher costs for long-distance links, enterprise-grade equipment, and maintenance, while often suffering higher latency and reliability issues compared with local area networks.

WANs require complex configuration, specialized design, and robust security controls, as broader exposure to external networks increases vulnerability to attacks, misconfigurations, and performance degradation.

WAN Real-World Uses

Corporate, Public, and Banking WANs

Corporate WANs link global offices and data centers. The public Internet connects millions worldwide. Banking WANs securely connect ATMs and branches to central banking systems.

Government, Cloud, and Remote Access WANs

Government and research WANs span countries. WANs connect campus LANs to cloud services. VPN-based remote access lets distant employees securely reach internal company resources.

05

Comparing LAN, WAN & PAN

LAN, WAN & PAN

Core traits

Coverage and scope
PAN connects devices around one person, LAN covers a room or campus, and WAN links cities, regions, or countries.

Speed and performance
PAN usually offers low to medium speeds, LANs provide high data rates, while WAN speeds are medium and vary with technology.

Ownership and control
PAN is owned by an individual, LAN is managed by a home or organization, and WAN involves telecom providers plus large organizations.

VS

Scale effects

Cost and scaling
As networks grow from PAN to LAN to WAN, per user or per site costs generally increase due to wider coverage and infrastructure.

Complexity and security
Configuration, security, and management are simple in PAN, more involved in LAN, and highly complex in WAN environments.

Layered connectivity
PAN, LAN, and WAN often interconnect, forming layered network paths from personal devices to campus networks and global services.

Feature & Use-Case Summary Table

LAN, WAN, PAN Feature Comparison

LANs cover buildings or campuses with high speed and often private ownership; WANs span cities or globally; PANs serve very short-range, personal devices and connections.

Patterns and Best-Fit Scenarios

PANs suit personal device connectivity; LANs fit homes, labs, and organizations; WANs interconnect distant sites. Recognizing coverage, speed, and ownership differences guides appropriate network type selection.

Type	Coverage Area	Speed	Typical Technologies	Ownership	Common Uses
PAN (Personal Area Network)	Within a few meters around a person	Low to moderate	Bluetooth, NFC, Zigbee, USB tethering	Individual user	Wearables,
LAN (Local Area Network)	Room, building, or campus	High	Ethernet, Wi-Fi (WLAN), switches, routers	Usually private (home or organization)	Home/office networking, labs, campus networks
WAN (Wide Area Network)	City, country, or global	Moderate to high (but often slower than LAN)	MPLS, leased lines, fiber backbone, 4G/5G, satellite	Mix of telecom providers and large organizations	Connecting branch offices, internet backbone, inter-city/

Visual Integration: How PAN, LAN & WAN Connect

Stepwise flow: PAN to LAN connection

Personal devices form a PAN around the user, linking wearables, peripherals, and sensors to a smartphone or laptop, which then joins a LAN via home or office Wi-Fi.

Stepwise flow: LAN to WAN and beyond

The LAN reaches the WAN through routers and ISPs, enabling access to remote websites, cloud platforms and corporate data centers in a continuous back-and-forth data flow.

Conclusion & Best Practices

Key takeaways and comparisons

PAN links personal devices, LAN connects local users, WAN joins distant sites. Each differs in range, cost, speed, management complexity, and typical advantages or limitations.

Practical choices, security, and next steps

Use PAN for wearables and phones, LAN for homes or campuses, WAN for multi-site enterprises. Apply strong passwords, encryption, updates; explore IP addressing, routing, security, wireless standards.

Thanks

Expressing Gratitude and Appreciation

Presenter: Rucha Soni